Functions in the form (ax + b) to a po

1 Find the following indefinite integrals, using C as the constant of integration:

a $\int (x-5)^4\,dx$

b $\int \dfrac{1}{(x+6)^3}dx$

c $\int 6\,(x+2)^2\,dx$

d $\int \dfrac{4}{(x-5)^3}dx$

e $\int (2x+3)^3\,dx$

f $\int (4x-3)^{-4}\,dx$

g $\int \dfrac{1}{(2x-3)^4}dx$

h $\int \dfrac{1}{(4x+3)^2}dx$

i $\int (2x-5)^{\frac{1}{2}}\,dx$

j $\int (4x+5)^{-\frac{1}{2}}\,dx$

k $\int (4x+5)^{\frac{5}{4}}\,dx$

l $\int (1-x)^6\,dx$

m $\int 4\,(5-x)^3\,dx$

n $\int (4-3x)^6\,dx$

o $\int 4\,(2x+5)^3\,dx$

p $\int 4\,(2-3x)^3\,dx$

q $\int (4-3x)^{-5}\,dx$

r $\int \dfrac{5}{(5-4x)^3}dx$

s $\int \sqrt{7-4x}\,dx$

t $\int 3\sqrt{2x+3}\,dx$

u $\int \dfrac{5}{\sqrt{1-4x}}dx$

v $\int 16\sqrt[3]{6x+5}\,dx$

w $\int \left(5+(4x+1)^3\right)dx$

x $\int \dfrac{5}{\sqrt{x-2}}dx$

y $\int \left(\sqrt{x-2}+\sqrt{x+3}\right)dx$

2 Find the equation of y in terms of x, given that $\dfrac{dy}{dx}=12\,(2x-1)^3$ and $y=1$ when $x=1$.

3 Find the equation of the curve that has a gradient function $\dfrac{dy}{dx}=5\,(5x-6)^2$ and the point $(1,1)$ lies on the curve.

4 For each of the following gradient functions, find the equation of y:

a $\dfrac{dy}{dx}=\sqrt{6x+66}$ and function y passes through the point $(-5,17)$.

b $\dfrac{dy}{dx}=(4x-2)^3$ and function y passes through the point $(2,79)$.

Functions in the form f'(x).f(x)

5 Consider the function $\left(x^2-3\right)^5$.

a Calculate $\dfrac{d}{dx}\left(x^2-3\right)^5$.

b Hence find $\int 30x\left(x^2-3\right)^4 dx$.

6 Consider the function $\left(5x^2+10x-3\right)^5$.

a Calculate $\dfrac{d}{dx}\left(5x^2+10x-3\right)^5$.

b Find $\int 150(x+1)\left(5x^2+10x-3\right)^4 dx$.

7 Consider the function $\left(x^2-3x+6\right)^5$.

a Calculate $\dfrac{d}{dx}\left(x^2-3x+6\right)^5$.

b Hence find $\int (3-2x)\left(x^2-3x+6\right)^4 dx$.

8 Consider the function $\sqrt{4x+11}$.

a Calculate $\dfrac{d}{dx}\left(\sqrt{4x+11}\right)$

b Hence find $\int \dfrac{12}{\sqrt{4x+11}}dx$.

9 Consider the function $\dfrac{1}{\left(x^2+7\right)^3}$.

a Calculate $\dfrac{d}{dx}\left(\dfrac{1}{\left(x^2+7\right)^3}\right)$.

b Hence find $\int -\dfrac{24x}{\left(x^2+7\right)^4}dx$.

10 Consider the function $y=e^{x^3}$.

a Calculate $\dfrac{dy}{dx}$.

b Hence find $\int 3x^2e^{x^3}dx$.

11 Consider the function $y=e^{3x^4-5}$ and hence find $36\int x^3e^{3x^4-5}dx$.

12 Consider the function $y=\sin\left(x^5\right)$ and hence find $\int x^4\cos\left(x^5\right)dx$.

13 Consider the function $y=\cos\left(x^6+3x^5\right)$ and hence find $15\int\left(2x^5+5x^4\right)\sin\left(x^6+3x^5\right)dx$.

14 Consider the function $y = e^{\sqrt{x}}$ and hence [redacted] $\dfrac{4e^{\sqrt{x}}}{\sqrt{x}}\, dx$.

15 Find the following indefinite integrals:

a $\int 28\left(6x^3+1\right)\left(3x^4+2x\right)^6\, dx$

b $\int \sin 3x \cos 3x\, dx$

c $\int 2e^{\sin x} \cos x\, dx$

d $\int (x+2)e^{3x^2+12x}\, dx$

Integration as reverse of differentiation

16 Consider the function $y = \dfrac{\cos x}{e^x}$.

a Calculate $\dfrac{dy}{dx}$.

b Hence find $\int \dfrac{\sin x + \cos x}{e^x} dx$.

17 Consider the function $y = e^x \sin x$.

a Calculate $\dfrac{dy}{dx}$.

b Hence find $\int 3e^x(\sin x + \cos x)\, dx$.

18 Consider the function $y = \tan x$.

a Calculate $\dfrac{dy}{dx}$ using $\tan x = \dfrac{\sin x}{\cos x}$.

b Hence find $\int \dfrac{5}{\cos^2 x} dx$.

Rearrangement method

19 Given that $\sin 3t = 3\sin t - 4\sin^3(t)$

a Make $4\sin^3(t)$ the subject.

b Hence find $\int 3\sin^3(t)\, dt$.

20 Consider the function $y = xe^x$.

a Calculate $\dfrac{dy}{dx}$.

b Hence show that $xe^x = \int xe^x\, dx + \int e^x\, dx$.

c Rearrange the expression in part (b) to find $\int xe^x\, dx$.

21 Consider the function $y = \frac{x}{e^x}$.

a Calculate $\frac{dy}{dx}$.

b Hence show that $\frac{x}{e^x} = \int \frac{1}{e^x}\, dx - \int \frac{x}{e^x}\, dx$.

c Rearrange the expression in part (b) to find $\int \frac{x}{e^x}\, dx$.

22 Consider the function $y = x \cos x$.

a Calculate $\frac{dy}{dx}$.

b Use the rearrangement method to find $\int x \sin x \, dx$.